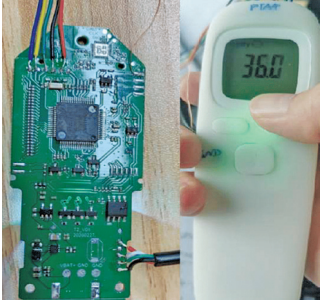
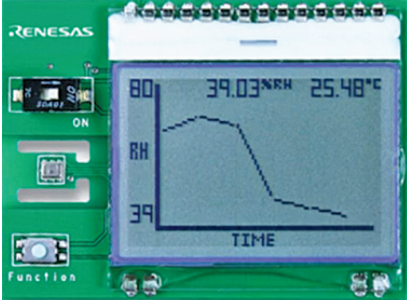


Interesting

Reference Designs OF THERMOMETERS

Title of Reference Design	Forehead Thermometer	IR Contactless Thermometer
Image		
Key Attractions	The reference design for the forehead thermometer combines low power consumption with high efficiency. Key features include a precise 16-bit ADC for accurate temperature readings and versatile connectivity options, making it ideal for advanced, battery-operated health devices.	The IR contactless thermometer design combines a high-precision MCU, efficient lithium-ion charger, and reliable LDO in one solution, optimising healthcare device development with integrated charging and stable power, making it perfect for critical applications.
Highlights	A forehead thermometer reference design from NXP Semiconductors uses the K32L2B microcontroller, designed for low-power, battery-operated applications. The design includes a 48MHz ARM Cortex-M0+ core with 256k flash memory, aiming for efficiency. An important part of the design is the integrated multi-channel 16-bit analogue-to-digital converter (ADC), necessary for processing signals from the infrared temperature sensor and measuring battery voltage. The MCU's ADC provides accuracy, with 13.9 bits in single-ended mode and 14.5 bits in differential mode, allowing the thermometer to measure temperature with precision up to 0.1 degrees. The design includes a 12-bit DAC to supply voltage to external amplifiers, removing the need for an external chip and offering a 1.2V reference voltage with a 1mA load capacity. It features an SLCD display that supports 24x8 or 28x4 segments, with pins that can function as segment and common ports or for other IO functions. Connectivity is expanded with three I2C channels for connections to sensors and devices, including an infrared sensor, a digital temperature sensor like the PCT2075, and flexible communications via FlexIO for I2C, UART, or SPI. PWM is used for LED indicators or buzzer alarms and supports voice playback.	Renesas' IR contactless thermometer reference design features the RL78/H1D microcontroller, ISL9205D lithium-ion battery charger, and ISL9021A low dropout regulator (LDO), offering a unified solution for IR contactless thermometer development. The ISL9205D provides a cost-effective, integrated charging solution for single-cell lithium-ion batteries with high input voltage capability, and the high-performance LDO ensures a stable and reliable power supply, enhancing the design's overall efficiency. This design features a high-precision analogue MCU at its heart, performing multiple functions such as signal amplification, system control, and LCD driving within a single chip, enhanced by the integration of a high-performance charger and LDO capable of managing high input voltages for a comprehensive and efficient solution. Tailored for sensing transducers in healthcare devices, particularly optimised for blood pressure monitors, the MCU is ideal for critical healthcare applications, supporting a voltage range from 1.8V to 5.5V and offering up to 8kB RAM and 128kB program flash. It includes diverse on-chip oscillator frequencies from 1 to 24MHz and a low-speed option at 15kHz, along with an RTC, power-on reset, and low voltage detection features.
Applications	The reference designs are suitable for cost-sensitive and low-power battery-operated applications.	They are ideal for critical healthcare applications and devices.
OEM Brand	NXP Semiconductors	Renesas
URL	https://www.electronicsforu.com/electronics-projects/reference-design-of-forehead-thermometer	https://www.electronicsforu.com/electronics-projects/ir-contactless-thermometer-reference-design